

SUAS 2026 Technical Design Report

Alex Eagles

Alexandria University

Alexandria, Egypt

alex_eagles@alexu.edu.eg

Abstract—This paper presents the design, engineering, and testing of a multirotor unmanned aerial system for the 2026 Student Unmanned Aerial Systems (SUAS) competition. Learning from a 2025 platform that was heavier than the mission required, this year’s strategy prioritizes reliability over capability: a smaller, lighter aircraft that completes its flight tasks reliably in one flight while scoring the transport and operator tasks by design. A weighted trade study selected an X-quadcopter, and propeller-driven sizing set the frame geometry, giving a 6.8 kg aircraft that meets the full-points weight threshold and collapses to a carry-on footprint. Image processing was moved from an onboard computer to a custom ground control station, cutting onboard weight and heat while allowing a larger detection model. The system was validated through staged testing spanning component and sensor tests, mission simulation, control tuning, and full flight rehearsals, with mapping flights setting the final capture configuration. Each choice traces back to one goal: reliably completing the highest-value tasks across the mission.

Index Terms—UAV, SUAS 2026, multirotor, autonomous flight, computer vision, payload delivery, aerial mapping

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We are grateful for the generous support of our sponsors: Hitec supplied servo motors free of charge and advised on servo selection for our payload delivery mechanism; T-Motor discounted our propulsion components, including the MN6007II-X motor set; RJX Hobby and Easy Composites discounted our foldable hinge hardware and carbon fiber sheets, respectively; and JLCPCB discounted PCB manufacturing for our custom power distribution board.

MathWorks, Altium, and eCalc [10] generously provided free software licenses used throughout our development, with Altium’s license further including a training course that supported our PCB design and thermal simulation work.

I. COMPETITION STRATEGY

A. Strategic Vision

Our goal this year is a multirotor built around reliability, where every task we attempt works consistently in a single flight.

Competing at SUAS 2025 exposed two problems with our platform: an oversized, reused propulsion system, and excess capability beyond what the mission required. This year’s design prioritizes a smaller, lighter airframe, which simplifies PID tuning through reduced inertia, lowers build and maintenance cost, and extends endurance enough to complete waypoint laps, mapping, and Search, Detect, and Deliver in a single flight. All-up weight fell from roughly 18 kg last year to 6.8 kg, a reduction of over 60%.

B. Airframe Configuration

Airframe selection was driven by a single mission requirement: precise, stationary payload release over a fixed target. A fixed-wing platform cannot hold a precise hover for the drop, and autonomous landing remains its highest-risk phase. A VTOL hovers but adds complexity and cost beyond our budget. A multirotor was selected for hover stability, tight turn radius, and low-risk autonomous takeoff and landing.

Table I summarizes the configuration trade study. Criteria weights were derived from the 2026 mission scoring distribution [8], with manufacturing cost and pilot training added to reflect budget and team experience constraints. The quadcopter scored highest at 77.4.

TABLE I
AIRCRAFT CONFIGURATION EVALUATION

Criteria	Wt(%)	FW	Quad	Hexa	VTOL
Endurance	20	0.90	0.50	0.40	0.70
Auto TO/Landing	5	0.30	1.00	1.00	0.60
Airdrop Precision	20	0.10	0.90	0.90	0.85
Portability	12	0.20	0.90	0.70	0.40
Weight	8	0.90	0.60	0.40	0.30
Imaging Stability	15	0.40	0.80	0.90	0.70
Manufacturing Cost	12	0.70	0.80	0.50	0.20
Pilot Training	8	0.30	0.90	0.80	0.50
Total	100	47.9	77.4	68.5	58.1

C. Mission Flow Strategy

Mission Time is 45 minutes, covering setup, flight, and deliverables. All waypoint laps must be flown before entering the Search Boundary, where the lap count locks. Because the camera captures geotagged imagery throughout the laps, and the lap path often passes near a Search Boundary, we typically enter Search, Detect, and Deliver already holding candidate frames of the search area rather than starting from nothing: the detection model returns candidates, and the operator verifies each one's location within the correct boundary and identity as a genuine tent or mannequin before any drop.

After the endurance laps, the vehicle surveys the search area in a lawnmower pattern for Risk Mapping, capturing imagery that also feeds the detection model. By the end of the pass we expect both objects detected and localized, so the vehicle proceeds directly to each drop location and releases the matching object.

Endurance waypoints are not published before Check-In, so we sized Risk Mapping and Search, Detect, and Deliver first and allocate the remaining Mission Time to endurance laps. SITL simulation over the two published Search Boundaries set a survey speed of 6 m/s, below lap cruise speed, to preserve the image overlap and sharpness that Risk Mapping's map-quality score depends on, covering the area in 10 minutes. Search, Detect, and Deliver adds roughly 2 minutes per object, about 4 minutes total. Together these account for about 15 minutes, leaving 15–20 minutes for endurance laps depending on setup time, bounded by the battery reserve held for landing. Since en-

durance scores as $200 \times (N_L/10)^2$, this remaining time, not lap quality, sets the final score.

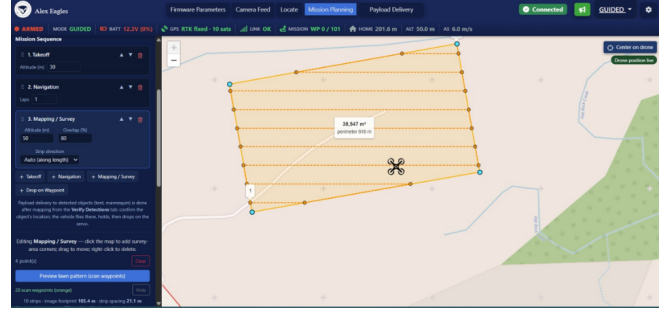


Fig. 1. Mapping Strategy

D. Complexity versus Reliability Trade-offs

Onboard companion computer. Last year's aircraft carried an onboard computer for image processing. Moving processing to the Ground Control Station removed the cooling, power, and weight constraints of an embedded system. It also saved hours of hardware integration and troubleshooting, which we spent on flight testing instead. The device running our GCS can support a heavier detection model than our onboard unit could. The cost is a dependency on the RF downlink. We accepted this because the pipeline works on stored geotagged stills rather than continuous video, which reduces bandwidth load and lets us batch images per inference call.

Payload delivery mechanism. For our payload delivery mechanism, we chose a simple, passive, and consistent design over a highly accurate, controlled one favoring easy setup and manufacturing, and avoiding adding unnecessary features like a DC motor power supply and motor driver.

Validation platform. We build and fly the full system on a low-cost wooden platform before committing to the final airframe. Integration and control problems appear on a frame we can afford to damage, not on the competition vehicle late in the schedule.

Integration approach. The aircraft is developed as one system, not a set of independent modules. Mechanical, electrical, and software interfaces were treated as design drivers from the start.

E. Efficient Operators

In line with our focus on reliability over added complexity, the system runs with two people: a safety pilot for manual override, and a GCS operator who monitors the mission. This lean crew is a direct result of the autonomy built into the system, scoring the full 200 points under the operator-count formula.

F. Design for Transport

The airframe collapses to a Carry-On footprint of 22 in $\times$ 14 in $\times$ 9.0 in (559 $\times$ 356 $\times$ 229 mm) and unpacks within the three-minute limit (Fig. 2).

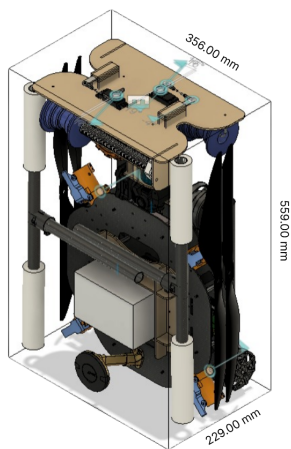


Fig. 2. Carry on footprint of 22 $\times$ 14 $\times$ 9.0 in.



Fig. 3. Neith Alex Eagle's UAS for SUAS 2026

All other components, including the battery, stay mounted in the collapsed state, so unpacking is limited to three connections: legs (tool-less quick-release), arms and payload mechanism

(keyed pins and nuts, one wrench), and camera (hand-mounted into its vibration dampers). With everything else pre-installed, deployment stays fast despite the single wrench step.

We chose the Carry-On tier over the smaller Personal Item tier to maximize aircraft capability while keeping deployment fast. Tool-less fasteners would score additional points, but our keyed-pin connections offer greater structural reliability and repeatable assembly at only a minor time cost.

All-up weight (AUW) is 6.8 kg, against the 35 lb competition limit and the 15 lb threshold for full weight points. The two delivery objects contribute about 410 g; the main driver of this year's weight reduction was downsizing the propulsion system from last year's configuration.

II. DESIGN STRATEGY

A. Design Methodology

Our design strategy follows from the competition strategy: build a smaller, lighter, more reliable aircraft than last year, and justify each choice against a measurable requirement. We turned the rulebook and our 2025 experience into concrete targets for weight, endurance, and accuracy, then sized the airframe, propulsion, and power systems to meet them. Work was divided among sub-teams for airframe, propulsion, software, hardware, and computer vision, each with an integration lead who checked decisions against the rest of the system before they were committed, so interface problems surfaced during design rather than assembly.

B. Airframe Design

1) *Frame Sizing*: Multicopter size is governed by propeller radius. For n arms spaced at $\theta = 360^\circ/n$, airframe radius R relates to the maximum propeller radius $r_{\max}$ as [1]

$$R = \frac{r_{\max}}{\sin(\theta/2)} = \frac{r_{\max}}{\sin(180^\circ/n)} \quad (1)$$

To avoid thrust loss from aerodynamic interference between adjacent rotors, the recommended clearance margin is [1]

$$r_{\max} = 1.05 r_p \text{ to } 1.2 r_p \quad (2)$$

For the MF2211 propeller ($r_p = 284$ mm), a mid-range margin of 1.12 gives $r_{\max} \approx 318$ mm and $R \approx 450$ mm, setting a 900 mm minimum motor-to-motor diagonal. The geometry is shown in Fig. 4. The frame was built at 1000 mm for extra tip clearance and space for the central electronics bay.

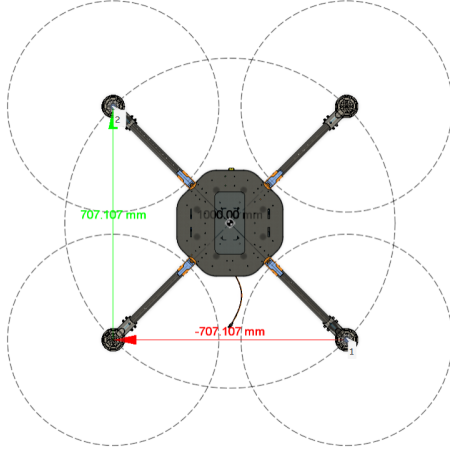


Fig. 4. Airframe radius derived from propeller radius.

2) *Structure and Materials*: The prototype used plywood for low cost and fast iteration; the final airframe uses carbon fibre for its higher stiffness-to-weight ratio, which also shifts structural natural frequencies away from motor excitation, reducing vibration at the flight controller.

The central structure is two CNC-cut carbon-fibre plates. The 3 mm bottom plate carries all major loads – components, release mechanism, landing gear, and arm thrust – while the 2 mm top plate carries only the battery. Its access hole is closed with a wooden insert, saving mass where carbon fibre would add none. Both thicknesses were confirmed by the structural analysis in Section II-B4.

Each arm is a 25 mm carbon-fibre tube on an aluminium folding joint, with two T-shaped landing legs on quick-release hinges giving 150 mm of ground clearance for the payload mechanism.

3) *Integration and Layout*: Components are arranged on three levels: the release mechanism below the bottom plate; flight controller, air unit, telemetry, and power distribution on the middle level; and battery, GPS, and rangefinder on top, following the component choices in Section II-D.

EMI-sensitive modules are kept clear of the battery and power lines, heat-generating components are spaced for convective cooling, and linked components are grouped together to shorten wiring and limit the impact of any single harness fault. This layout also lets components be replaced without disassembling the aircraft, shortening turnaround between test flights.

4) *Structural Analysis*: The bottom plate was analysed in Autodesk Fusion under static loads at the four arm mounts. At the applied load (4.32 N/mount), maximum von Mises stress was 12.4 MPa against a 300 MPa yield strength, with 0.083 mm deflection. Scaled to takeoff thrust ($T/W = 2.3$, 39.5 N/mount), stress rises to 113 MPa – a safety factor of 2.7 – with 0.76 mm deflection (Fig. 5).

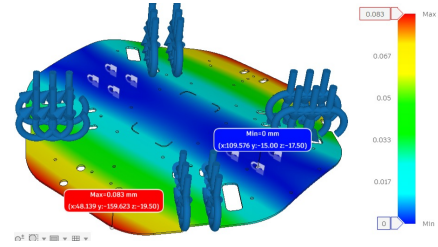


Fig. 5. Deflection of the bottom plate under the applied static load case.

C. Mass Breakdown

TABLE II
COMPONENT MASS BREAKDOWN

Group	Component	Mass (g)
Structure	Frame, arms, legs, hinges	1252
Power	Battery, PDB, wiring	2795
Avionics	FC, GPS, sensors, camera, Herelink Air Unit	483
Mission	Payload release mechanism	525
Propulsion	Motors, ESCs, props, mounts	1320
Dry Mass		6375
Payload	Water bottle + beacon	410
All-Up Weight		6785

Table II gives the mass breakdown. All-up weight, including both delivery objects, is 6.8 kg (15.0 lb), within the 15 lb threshold for full weight-scoring points. The battery is the largest single item, so it is placed at the geometric centre, with four identical, equally spaced arms keeping

the airframe symmetric about both axes. The flight controller is also centred at the centre of rotation, so its IMU does not mistake centripetal acceleration for linear acceleration during manoeuvres. The CAD model shows the loaded centre of gravity is 1.27 mm from the geometric centre, which is negligible at this scale.

D. Component Selection

Components were selected to meet concrete mission thresholds at the lowest integration cost, and Fig. 6 shows the system layout.

The Pixhawk Cube Orange+ flight controller was chosen over single-IMU F7/H7 boards for its triple-redundant IMU set and native ardupilot support. The Here3+ GPS was selected for its RTK integration – offering sub-meter accuracy required for payload mission– and DroneCAN interface. The Herelink communication system consolidates telemetry, RC control, and video into one low-latency link, integrating directly with the onboard camera over Ethernet; this concentrates risk on a single path but simplifies wiring and ground-station management. The LightWare SF45 lidar provides obstacle detection, chosen for its wide FOV and lightweight. The SIYI A8 Mini was selected for its built-in gimbal and native Herelink integration; since all image processing runs on the ground, no onboard companion computer is required, cutting power draw, weight, and thermal load.

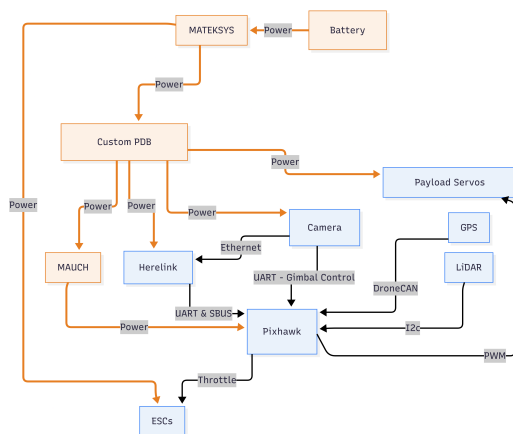


Fig. 6. Wiring architecture of UAV system.

E. Power Distribution

The power network uses a three-board architecture. A MATEKSYS HEX PDB distributes

primary battery power to all four ESCs and to a custom-designed PDB, built in-house to accommodate our specific payload and communications load profile on a single board reusable across platforms. Switching-mode power supplies and LDOs provide stable rails for all components (Table III). Thermal simulations produced current-density and voltage-drop heatmaps (Figs. 7 and 8) that guided layout decisions before fabrication. Flight controller power was isolated onto its own regulator, given the Pixhawk’s role as the aircraft’s single point of failure; a MAUCH BEC provides two independently regulated outputs, so either can fail without a reset or brownout.

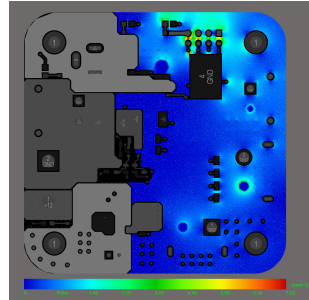


Fig. 7. Input power plane

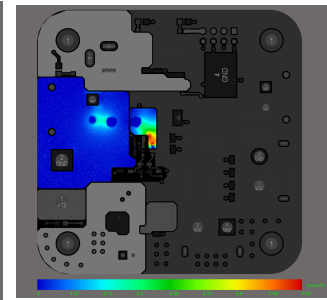


Fig. 8. Herelink power supply

TABLE III
CUSTOM POWER DISTRIBUTION PCB – COMPONENT POWER SUMMARY

Component	Regulation	Voltage
Herelink	SMPS + LDO (redundant)	10V
Servo	Buck converter	5V
DC Motor	Buck converter	12V
RFD900x	LDO	5V
Camera	Unregulated pass-through	Vbat
MAUCH BEC	Unregulated pass-through	Vbat

F. Payload Delivery

The payload deployment mechanism is a passive two-stage descent system, consisting of a servo-actuated release, a 3D-printed lead screw, a custom spool-nut assembly, a support shaft with bearings, and a rotary damper (Fig. 9).

The payload hangs beneath the UAV, retained by a servo-locked rope wound around the spool of a custom nut that engages the lead screw and stores the descent rope. On activation, the servo

releases the payload: the rope unwinds, driving the spool-nut axially along the stationary lead screw as the payload descends. When the spool-nut reaches the end stop, axial motion converts to rotational motion – the spool-nut and lead screw rotate together, unwinding the remaining rope at a controlled rate.

Bearings on the lead screw carry the radial and axial loads, while a rotary damper coupled to the screw shaft dissipates rotational energy to limit descent velocity. Since the bearings carry the structural loads, the damper sees only torsional loading, supporting reliable, long-service-life operation.

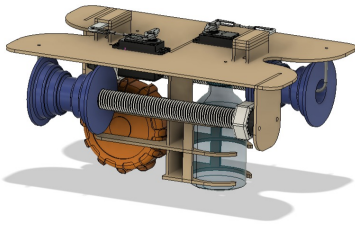


Fig. 9. Payload Dropping Mechanism Assembly.

G. Ground Control Station and Mission Software

Design philosophy. Most competition tasks need only straightforward autonomous navigation, so our software work went toward the two areas that shape the outcome: fast, low-error mission setup and a reliable operator check for Search, Detect, and Deliver. We build on a GCS developed over previous seasons and flown in earlier competitions, so this year extended a proven system rather than rebuilding one.

Architecture. The GCS is a web application with a React interface (Fig. 10) and a backend of independent services behind an API gateway: a firmware controller built on pymavlink, a camera service using the SIYI SDK, a detection service, and a delivery service. This architecture was validated end-to-end in SITL before flight (Sec. III).

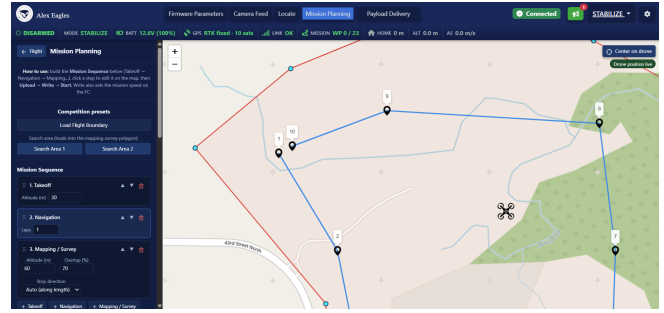


Fig. 10. Custom Ground Control Station interface.

One requirement shaped the interface before any task feature. The rules give the GCS judge continuous access to a display showing the flight boundaries, UAS position, ground speed, and AGL altitude. We keep this as a dedicated, always-on view, separate from the operator’s task tools, so a fault in one cannot hide the required display from the judge.

Mission setup. With that display fixed, the rest of the interface is built around saving time. The flight boundary and both search boundaries are pre-loaded and selected with a single action, and selecting a search boundary generates a lawnmower survey path automatically, in under three seconds for a 10-acre area. Endurance waypoints are issued at Check-In under time pressure, so the system accepts them by direct entry or file import, both tested in bench and SITL runs, to reduce transcription error.

Operator review. Figure 11 shows the full path from image capture to delivery command; the onboard capture, downlink, and batched-inference stages are detailed in Section II-H. Each detection reaches the operator as one candidate per object rather than per frame, following the k-means clustering [11] step described there. The operator accepts or rejects each candidate with a single action, and an accepted candidate becomes the drop location and object passed to the delivery mission. Limiting the operator to verification rather than search keeps the check within the time budget set in the Mission Flow Strategy.

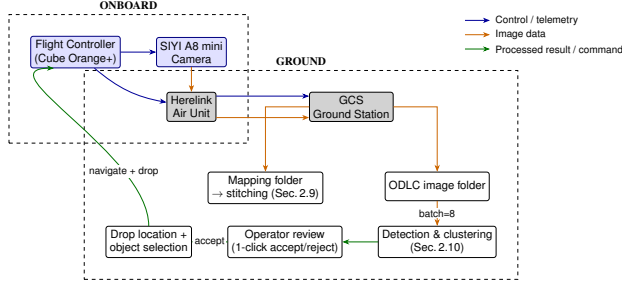


Fig. 11. Mission data architecture

H. Object Detection and Localization

Section I-D explained moving image processing off the aircraft and onto the GCS. This subsection covers the resulting pipeline: image capture and geotagging, the downlink and batching architecture, the detection model, and how a detected object’s pixel detection becomes a real-world delivery point.

1) *Image Acquisition and Geotagging:* A SIYI A8 mini camera connects via UART directly to the flight controller (Cube Orange+), which triggers capture and, since it holds the GPS solution, geotags each image at the moment of capture. This avoids adding a companion computer for triggering and tagging. Images are also saved to the camera’s SD card, giving a complete mission record independent of the RF link.

2) *Downlink and Batched Inference:* The camera bridges to the Herelink air unit over Ethernet, putting it on the same network as the GCS across the RF downlink. Two watcher processes manage image flow on the ground: one pulls newly captured images via the camera’s API, the other tracks accumulated image count. Once a multiple of batch size N is reached, the latest N images are sent to the detection model as one inference call, amortizing per-call latency across N frames. This is what allows a larger model on the GCS while keeping throughput high enough for the mission’s image volume, per the trade-off in Section I.

3) *Detection Model Selection:* We selected UAV-DETR [5], a real-time, end-to-end detection transformer built for aerial imagery, over generic real-time detectors (YOLO) and generic end-to-end detectors (RT-DETR) [6]. Both mission targets are small relative to the frame and often partly occluded by terrain, the exact failure mode

UAV-DETR targets. On VisDrone [7], a benchmark of small, dense, aerial-viewpoint objects, UAV-DETR outperforms comparable-size baselines (Table IV):

Model	Params (M)	GFLOPs	AP	AP ₅₀
YOLOv11-M	20.0	67.7	25.9	43.1
RT-DETR-R50	42	136	28.4	47.0
UAV-DETR-EV2	13	43	28.7	47.5
UAV-DETR-R18	20	77	29.8	48.8
UAV-DETR-R50	42	170	31.5	51.1

TABLE IV
DETECTION ACCURACY ON VISDRONE

We run the R50 variant, since GCS hardware carries none of an onboard computer’s power or weight constraints. Fig. 12 shows a sample detection from our pipeline, correctly identifying both a tent and a human target.



Fig. 12. UAV-DETR detections on a search-area image, showing the tent and human targets with confidence scores.

4) *Object Geolocation and Multi-Detection Resolution:* Each detection at pixel (u, v) , relative to image center $(u_{\text{center}}, v_{\text{center}})$, is projected to a ground offset in meters:

$$X = \left(\frac{u - u_{\text{center}}}{I_w} \right) W, \quad Y = \left(\frac{v - v_{\text{center}}}{I_h} \right) H,$$

where I_w, I_h are image dimensions in pixels and W, H are the ground footprint at altitude h :

$$W = 2h \tan \left(\frac{\text{FOV}_x}{2} \right), \quad H = 2h \tan \left(\frac{\text{FOV}_y}{2} \right).$$

$\text{FOV}_x = 81^\circ$ is the A8 mini’s published horizontal FOV [12]. Since SIYI publishes only a

diagonal FOV of 93° , we derive FOV_y from the rectilinear-lens relation

$$\tan^2\left(\frac{\text{FOV}_{\text{diag}}}{2}\right) = \tan^2\left(\frac{\text{FOV}_x}{2}\right) + \tan^2\left(\frac{\text{FOV}_y}{2}\right),$$

giving $\text{FOV}_y \approx 63^\circ$.

The offset (X, Y) converts to latitude/longitude using the aircraft's GPS position (ϕ_0, λ_0) at capture:

$$\phi = \phi_0 + \frac{Y}{111320}, \quad \lambda = \lambda_0 + \frac{X}{111320 \cos(\phi_0)},$$

with 111,320 m as one degree of latitude on WGS84 [9]. Each detection thus produces its own spatial estimate rather than using the aircraft's position as a stand-in.

Because the aircraft flies a lawnmower search pattern, the same object appears in several overlapping frames, giving multiple independent coordinate estimates per target. We resolve these with k -means clustering, fixing k to the known number of targets (e.g., two in the 2026 mission: one mannequin, one tent); each cluster's centroid is the object's final reported position. Since the projection step already accounts for the camera's footprint, clustering only averages out residual noise, e.g. GPS error, altitude variance, and gimbal deviation from nadir, reducing any single frame's impact on the final delivery point.

I. Mapping

The mapping pipeline processes the geotagged imagery captured by the SIYI A8 mini during the lawnmower search pattern described in Section II-H. To construct the final orthomosaic, we use OpenDroneMap [4] (ODM) rather than a conventional feature-matching image stitcher, balancing processing requirements with global map accuracy.

This choice addresses a structural limitation of naive stitching over large competition areas: a conventional stitcher matches pixel features between neighboring image pairs only, and with no external knowledge of camera position, small alignment errors compound across frames, often causing severe geometric distortion and map bowing. ODM avoids this by using the flight controller's GPS geotags as positional priors, constraining feature-matching to true spatial coordinates and producing a globally consistent, georeferenced map.

III. TESTING STRATEGY

A. Testing Approach and Plan

Our testing follows a build-up approach: components are validated together on a low-cost development platform first, then on the integrated aircraft, and finally in full autonomous flight. Finding a fault at the lowest level of integration keeps it cheap to fix and easy to trace, before it can propagate into the complete system. We built and flew a development platform: a 4.5 kg wooden quadcopter using T-Motor MN5008 motors, built close to the final competition specification and chosen so that damage during early iteration would not be costly (Fig. 13). Its purpose was to validate that our components worked together as an integrated system, and to test the PDB design against a configuration near the aircraft's real load and layout before the final airframe was fabricated.

1) *Test Plan*: Testing followed four progressive stages, each gated by the acceptance criteria in Table V.

TABLE V
TEST PLAN STAGES AND ACCEPTANCE CRITERIA

Stage	Acceptance Criterion
Component – hardware validated in isolation	Each unit powers and operates within its rated limits.
Sensor / control loop – tuning on the development platform	Stable attitude tracking and clean sensor logs after tuning.
Subsystem – navigation, perception, communication, payload	Each meets its own target independently.
Full mission – staged autonomous rehearsals	Complete mission profile flown end to end.



Fig. 13. Wooden quadcopter development platform (4.5 kg, MN5008 motors) used for integrated component testing.

B. Component and Sensor Testing

Each component was tested against a defined objective before integration.

Component power distribution board: For the custom PDB, Load testing was performed on each regulator up to its maximum rated electrical capacity, confirming that all rails held within tolerance under sustained full load. Once verified, the PDB was tested on spare, non-critical components to confirm it delivered the correct voltage to each rail without damaging any load. It was then connected to the full component stack, powering the entire system. A thermal imaging camera was used to check for hotspots under high-ambient-temperature conditions, and the resulting thermal profile was compared against the simulation results in Section II-E. Measured hotspot locations and magnitudes matched the simulated current-density heatmaps, validating the simulation as a reliable design tool for future revisions.

Propulsion: Using a test rig equipped with a wattmeter, each motor was tested across a range of throttle levels and propeller combinations. Measured thrust and current draw were compared against the manufacturer’s performance data to validate efficiency, and the measured current draw was used to calculate a more realistic flight-time estimate than the manufacturer specifications alone provided.

GPS and RTK System: Positioning accuracy was validated by commanding the aircraft to hover over a marked ground target and measuring its position offset, and by sending it to known surveyed coordinates and measuring the resulting error. Without RTK corrections, positioning error measured 3–5 m; with RTK enabled, error was reduced to as low as 3 cm [9], confirming RTK as necessary for the sub-metre accuracy the payload delivery mission requires.

Camera: The imaging system uses a SIYI A8 Mini. We built a dedicated test interface to exercise gimbal control, the video stream, and the capture system independently (Fig. 14). The camera was first tested for geotagged capture, confirming each triggered image carried correct GPS and orientation data. We then compared image transfer to the ground station one-by-one against batched transfer; batched transfer moved images faster and was adopted for the mission pipeline.

SF45/B Lidar: Detection was checked against poles, trees, and moving obstacles at varying distances in both open fields and cluttered environments. The system reliably recognised obstacles across repeated trials.

Telemetry and video link. The Herelink air unit was range-tested by walking the ground unit progressively away from the aircraft while monitoring link quality, and by deliberately obstructing the antennas to observe behaviour under a degraded link. The link held telemetry, control, and video out to approximately 1.5 km, and obstructing the antennas produced no loss of control authority or video.

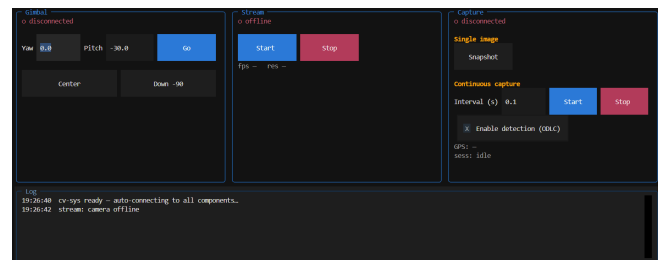


Fig. 14. Camera test interface for gimbal control, video stream, and geotagged capture.

C. Software and Mission Simulation

We test every mission in simulation before flying it. Using ArduPilot [2] Software-in-the-Loop (SITL) through Mission Planner [3], we run the full mission sequence against a simulated vehicle: the waypoints, the generated survey path, and the geofence limits. This confirms the aircraft follows the intended sequence before any real flight.

The same setup lets us test our custom GCS end to end without leaving the ground. Because SITL provides a simulated vehicle that behaves like the

real one, we connect the GCS to it and exercise the full operator workflow: uploading a mission, generating the survey path, streaming telemetry, and issuing commands such as takeoff, Return-to-Launch, and the payload drop. This lets us confirm the GCS sends valid, executable commands and displays the vehicle’s position and state correctly, so problems in the software are found and fixed in simulation rather than during a flight. A mission is only flown on the real aircraft after both the mission logic and the GCS pass in simulation.

D. Control-System Testing and Tuning

Flight control was brought up in stages, following the ArduPilot tuning methodology, with log analysis between every step. At each stage we reviewed vibration levels, magnetometer interference, GPS quality, and all sensor readings before advancing.

First flights were flown in Stabilize mode, applying light pitch and roll inputs to check response and reducing PID values if the aircraft did not handle well until an acceptable response was observed. This method was repeated for altitude hold, after which we applied a notch filter to reject motor-frequency vibration. Next, Loiter mode was used to perform manual and autotune, significantly improving response time, and reviewing logs after each to decide whether further tuning was needed. Basic autonomous functions—takeoff, return-to-launch, and guided mode—were verified before any full mission was attempted.

Figure 15 shows attitude tracking from a post-tune flight log, with mean roll and pitch each within 0.1° of the commanded value (roll: 4.14° vs. 4.18° ; pitch: 4.96° vs. 5.04°), confirming accurate attitude response after tuning.

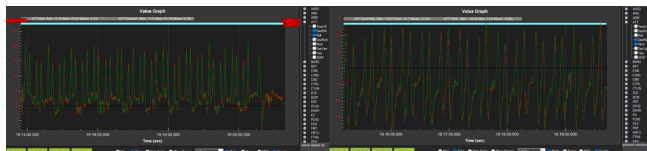


Fig. 15. Post-tune Pitch tracking

E. Flight and Mission Testing

Once tuning was complete and the aircraft flew reliably, we moved to full mission rehearsals.

Rather than testing the mission as one large event, we rehearsed each part on its own and then together: full waypoint navigation, the mapping mission, and the delivery mission. Running these repeatedly let us confirm the aircraft, GCS, and mission logic worked as one system under real flight conditions, and gave the operators practice with the mission timeline before competition.

1) *Mapping*: Mapping was tested across several flights, and the results set the capture configuration used in the mission. On early flights at a cruise speed of about 10 m/s, with the camera triggering every 0.3 s at 4K resolution, OpenDroneMap could not reconstruct a usable map. The camera could not write each 4K frame before the next trigger, so far fewer images were captured than commanded, leaving too little forward overlap. Reducing the survey speed to 6 m/s and lowering the resolution to 1080p let the camera keep up with the trigger interval, raising the number of accepted captures and the overlap between them. At this configuration OpenDroneMap reconstructed the map successfully, as shown in Fig. 16.



Fig. 16. Orthomosaic reconstructed from a flight survey at the tuned capture configuration (1080p, 6m/s).

REFERENCES

- [1] Q. Quan, *Introduction to Multicopter Design and Control*. Singapore: Springer, 2017.
- [2] ArduPilot Development Team, “ArduPilot Documentation.” [Online]. Available: <https://ardupilot.org/>
- [3] M. Osborne, “Mission Planner Ground Control Station.” [Online]. Available: <https://ardupilot.org/planner/>
- [4] OpenDroneMap Authors, “OpenDroneMap: A command line toolkit for processing aerial drone imagery.” [Online]. Available: <https://www.opendronemap.org/>
- [5] H. Wang et al., “UAV-DETR: Efficient End-to-End Object Detection for Unmanned Aerial Vehicle Imagery,” *arXiv preprint arXiv:2501.01855*, 2025.
- [6] Y. Zhao et al., “DETRs Beat YOLOs on Real-time Object Detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [7] P. Zhu et al., “Detection and Tracking Meet Drones Challenge,” *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 7380–7399, 2022.

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- [8] RoboNation, “Student Unmanned Aerial Systems (SUAS) 2026 Competition Rules,” 2026. [Online]. Available: <https://robonation.org/programs/suas/>
 - [9] P. J. G. Teunissen and O. Montenbruck, Eds., *Springer Handbook of Global Navigation Satellite Systems*. Cham: Springer, 2017.
 - [10] M. Müller, “eCalc – xcopterCalc: the most reliable multicopter calculator on the web.” [Online]. Available: <https://www.ecalc.ch/>
 - [11] J. MacQueen, “Some methods for classification and analysis of multivariate observations,” in *Proc. 5th Berkeley Symp. Mathematical Statistics and Probability*, 1967, pp. 281–297.
 - [12] SIYI Technology, “A8 mini Gimbal Camera User Manual,” 2024. [Online]. Available: <https://siyi.biz/>